PROJECT-BASED LEARNING

PHASE-II EVALUATION

Market Lens

Business Dashboard for Tracking Marketing Trends on Social Media Platforms

Team ID: AJP-III-2026-T053

Department of Computer Applications
Graphic Era (Deemed to be University), Dehradun
Academic Session 2025–2027

Team & Mentor Details

Team Information

Team ID: AJP-III-2026-T053
Semester: 3rd
Section: MCA(G4)
Domain: Web Development

Team Members

1. Suhani Gupta – 2520230432
2. Angel Goswami – 2520230407
3. Urvi Sharma – 2520230239

Mentor

Dr. Bhawnesh Kumar **Interactions completed:** 2

Our team has three members. We are working together to develop Market Lens with guidance from our mentor.

Phase-I Recap & Feedback Incorporated

Phase-I Status

During Phase-I, we planned Market Lens, selected the technologies and prepared the basic system design. The development work started after Phase-I.

Feedback Received

- Make the project explanation clear and original.
- Explain how the planned system will work.
- Show the actual development progress in Phase-II.

Changes Incorporated

- Prepared the project report and ppt using original content without copying.
- Developed the React dashboard.
- Set up Spring Boot and connected MySQL.

Problem Statement & Finalized Objectives

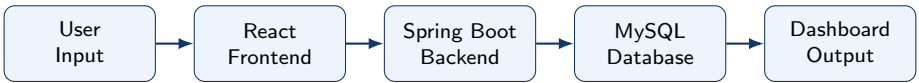
Problem Statement

Social media trends keep changing and it is difficult to track them across different platforms. Users often have to visit multiple websites to check popular topics, engagement and competitor activities. Market Lens aims to bring this information together in one dashboard.

Finalized Objectives

1. Build a simple dashboard to explore social media trends.
2. Store and manage competitor information using MySQL.
3. Add searching, sorting and filtering to find useful data quickly.
4. Show trends and engagement information in a clear and organized way.

Updated System Architecture / Workflow



Major Components

- React for the dashboard and user interface.
- Spring Boot for backend logic and REST APIs.
- MySQL for storing competitor details.

Changes Since Phase-I

- Started developing the planned architecture.
- Connected React with Spring Boot.
- Added MySQL database connectivity.

Module Design & Functional Requirements

Major Modules

1. Dashboard Module
2. Competitor Management Module
3. Backend and API Module
4. Database Management Module

Functional Requirements

- Display the main dashboard and its pages.
- Allow users to add, view and delete competitors.
- Send and receive data using REST APIs.
- Store and retrieve competitor details from MySQL.

Module Dependency

The React frontend sends user requests to Spring Boot through REST APIs. Spring Boot communicates with MySQL to save or retrieve information and sends the results back to the dashboard.

Technology Stack & Development Methodology

Technology Stack

- **Programming:** Java, JavaScript
- **Frontend:** React
- **Backend:** Spring Boot
- **Database:** MySQL
- **Development Tools:** VS Code, IntelliJ IDEA
- **Version Control:** GitHub (planned)

Development Methodology

1. **Requirement Analysis**
   Understanding the project needs.
2. **System Design**
   Planning the dashboard and database.
3. **Module Development**
   Building the frontend and backend.
4. **Integration**
   Connecting React, Spring Boot and MySQL.
5. **Testing and Validation**
   Checking the working features.

Implementation of PBL Course Concepts

Course	Concepts Implemented	Project Module
DSA	Basic data handling using Java collections	Backend
DBMS	MySQL tables, SQL queries and data storage	Competitor Management

Current Implementation

We use MySQL to save competitor details. Spring Boot handles requests and connects the dashboard with the database.

Planned DSA Implementation

We plan to add searching, sorting and hashing to improve how data is managed in Market Lens.

Development & Implementation Progress

Module	Status	Work Completed
React Dashboard	Mostly Completed	Created the dashboard and main pages.
Competitor Management	Completed	Added options to add, view and delete competitors.
Spring Boot & MySQL	In Progress	Connected the backend with the database and REST APIs.
DSA & Live Data	Pending	Searching, sorting and real social media data are planned.

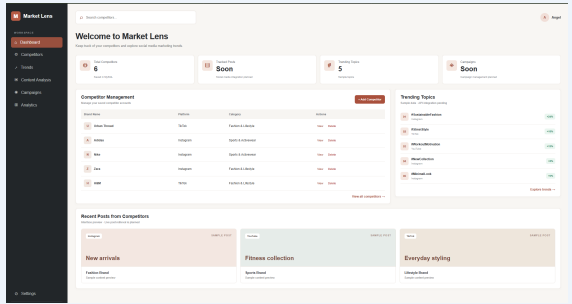
Overall Project Completion: Approximately 55%

Working Prototype / Demonstration

Implemented Features

- React dashboard with main pages.
- Add new competitor details.
- View saved competitors.
- Delete competitor records.
- Data stored in MySQL through Spring Boot.

Prototype / Output



Testing, Results & Validation

Test	Expected Result	Observed Result	Status
Add Competitor	Record should be saved.	Saved in MySQL.	Pass
View Competitors	Records should appear.	Shown on dashboard.	Pass
Delete Competitor	Record should be removed.	Deleted successfully.	Pass

Results / Validation

The add, view and delete features worked during testing. Competitor data was saved in MySQL and remained available after refreshing the dashboard.

GitHub Repository & Development Activity

Repository Information

- **Project:** Market Lens
- **GitHub:** T-053/MarketLens
- **README:** To be updated
- **Contributors:** Three team members
- **Branches:** Not created separately yet

Development Activity

- Created the React frontend.
- Developed the Spring Boot backend.
- Connected MySQL with Spring Boot.
- Added basic competitor management.
- GitHub activity is being organized.

Repository Link

`https://github.com/T-053/MarketLens/tree/main`

Individual Contribution

Member	Role	Technical Contribution	Commits
Suhani Gupta	Database Developer	Created the MySQL database and competitor table. Connected MySQL with Spring Boot.	–
Angel Goswami	Frontend and Backend	Developed the React dashboard, set up Spring Boot and added competitor management features.	–
Urvi Sharma	Testing and UI	Displayed competitor details on the dashboard and tested the add, view and delete features.	–

Team Contribution

We divided the project work among three members and worked together to connect and test the modules.

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- Connecting Spring Boot with MySQL.
- Sending data between React and Spring Boot.
- Fixing errors while saving and displaying data.

Solutions / Learning

- Fixed database connection errors.
- Learned how REST APIs connect frontend and backend.
- Tested the features to find and fix errors.

Phase-III Roadmap

1. Add searching, sorting and hashing.
2. Connect real social media data using APIs.
3. Add WebSockets for live updates.
4. Use Docker for deployment.
5. Complete testing and documentation.

Expected Outcomes & References

Expected Final Outcomes

- A working dashboard for social media trends.
- Easy management of competitor information.
- Searching and sorting of trend data.
- Live updates from social media platforms.
- A system that can be improved in the future.

References

1. React Documentation
   `https://react.dev`
2. Spring Boot Documentation
   `https://spring.io/projects/spring-boot`
3. MySQL Documentation
   `https://dev.mysql.com/doc/`
4. Docker Documentation
   `https://docs.docker.com`

Thank You